MICHELLE POLINKA
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SUMMARY

Full-stack software engineer with an M.S. in CS, currently building AI-powered systems. Experienced in LLM integration, RAG, production APIs, and real-time data systems. Proficient in Python, C++, React, TypeScript, and SQL.

EDUCATION

M.S. Computer Science and Engineering: University of California, Santa Cruz
- September 2025 – June 2026

B.S. Computer Engineering: University of California, Santa Cruz
- September 2022 – June 2025

EXPERIENCE

UCSC Faculty-Led Team – Fullstack AI Engineer: June 2026 – Present
- Develop an AI-powered decision-support system with a 5-person team using RAG and LLM tool-calling workflows
- Contribute to an architecture where a rules engine computes critical outputs and LLMs are limited to narration
- Design ingestion pipelines that transform unstructured web sources into schema-validated, versioned JSON
- Implement citation-grounded hybrid retrieval combining structured rule lookups with a semantic vector index
- Integrate LLM orchestration routing user queries to deterministic tools and citation-backed retrieval

Parlo – Software Engineer: September 2025 – February 2026
- Engineered a real-time Firebase/Firestore synchronization layer supporting 150+ concurrent active users
- Redesigned Firestore data models and batch write workflows to eliminate race conditions across clients
- Optimized database query patterns and indexing strategies, contributing to a 17% increase in daily active users
- Built internal tooling and monitoring solutions to identify system bottlenecks and accelerate debugging
- Designed scalable backend architectures supporting rapid feature development, reliability, and performance

BiometerPlus – Software Engineer Intern: June 2024 – September 2024
- Designed RESTful API endpoints with MySQL integrations supporting biometric data collection and retrieval
- Built MySQL-backed authentication and account management systems validated through Postman test suites
- Implemented full account lifecycle management with soft deletion, reactivation, and JSON schema validation
- Developed a secure password recovery workflow using time-limited tokens and server-side validation checks
- Enhanced backend reliability through input validation, authentication controls, and structured error handling

PROJECTS

JazzyFS: Adaptive Experimental Filesystem (Master's Thesis)
- Designed and implemented a FUSE-based filesystem with adaptive file prefetching and runtime access analysis
- Built instrumentation and benchmarking tools to evaluate filesystem performance across multiple workload types
- Developed a sonification system converting filesystem activity into real-time audio feedback
- Created visualizations of runtime filesystem access patterns to support performance analysis
- Created automated experiment pipelines for repeatable testing, performance analysis, and result generation

Fault-Tolerant Distributed Sensor Network
- Developed a fault-tolerant token-ring communication system across three Raspberry Pi devices
- Built a MySQL-based data storage system for distributed sensor data collection and management
- Created automated monitoring tools to detect inactive devices across the distributed network
- Built alerting mechanisms notifying operators of device failures to improve system reliability
- Implemented communication and data transfer systems for reliable cross-device synchronization

TECHNICAL SKILLS

- **Languages:** Python, C++, TypeScript, SQL
- **Frameworks and Libraries:** React, FastAPI, FUSE
- AI/ML: LLM integration and tool-calling, retrieval-augmented generation (RAG), vector databases
- **Databases:** PostgreSQL, MySQL, Firebase/Firestore
- **Tools and Infrastructure:** Linux, Docker, Git, GitHub Actions, Bash, Postman

ADDITIONAL

- Advanced Certificate of Merit in Classical Piano
- Composed and copyrighted original music